Year 9 · Chapter 10 · Surface area and volume: answers (teacher copy)

10.6 Circumference and arc length

(booklet p. 1)

WE DO examples (teacher)

1. Ex 1a: C = 2 × π × 5 = 31.4 cm
2. Ex 1b: C = π × 12 = 37.7 m
3. Ex 1c: P = ½ × π × 10 + 10 = 25.7 cm
4. Ex 1d: P = ¼ × 2 × π × 6 + 6 + 6 = 21.4 m
5. Ex 2a: l = 60/360 × 2 × π × 9 = 9.4 cm
6. Ex 2b: P = 9.42… + 9 + 9 = 27.4 cm
7. Ex 2c: P = 10 + 6 + 10 + ½ × π × 6 = 35.4 m (not the dashed side)
8. Ex 2d: 50 = 2 × π × r, so r = 50 ÷ (2 × π) = 8.0 cm

Easy

1. 8 cm
2. 5 m
3. 14 m
4. 4.5 cm
5. 18.8 cm
6. 31.4 m
7. 7.9 m
8. 44.0 cm
9. 12.6 cm
10. 50.3 m
11. 9.4 m
12. 25.1 cm
13. 28.3 m
14. 75.4 mm
15. 20.6 cm
16. 30.8 m
17. 25.7 m
18. 14.3 cm
19. 35.7 m
20. 21.4 cm

Medium

1. 6.3 m
2. 15 m² (10.5)
3. 7.9 cm
4. 24.6 cm
5. 28.6 cm
6. 25.9 cm
7. 4.45 m to 4.55 m (10.2)
8. 3.0 m
9. 2.5% (10.3)

Worded problems

1. C = π × 3.2 = 10.05 m; 10.05 × $12 = $121
2. C = 2 × π × 33 = 207.3 cm; 100 000 ÷ 207.3 = 482.3, so 482 full turns
3. 84 + 84 + π × 64 = 369.1, so 369 m